Euphoria Solutions

Euphoria Solutions is a newly established software development and IT solutions company that provides custom software development, web applications, business automation systems, technical support, and cloud-based solutions for small and medium-sized businesses.

The company initially employs 20 personnel divided into four major departments: Information Technology, Human Resources, Finance, and Sales. Since the company is starting with no existing IT infrastructure, a complete infrastructure plan is required before equipment and services are purchased.

Company Vision

To become a trusted technology partner for businesses by providing reliable, secure, innovative, and affordable software and IT solutions.

Company Mission

Euphoria Solutions aims to develop practical technology solutions that improve business operations, productivity, communication, and customer service. The company is committed to providing quality software, dependable technical support, and secure IT services while continuously improving its technology and processes.

Office Location

2rd Floor, Euphoria Business Center,

1957 Juan Luna Street, Tondo Manila

The company occupies a single office floor with dedicated work areas for the Information Technology, Human Resources, Finance, and Sales departments.

Organizational Structure

General Manager

- Information Technology Department
- Human Resources Department
- Finance Department
- Sales Department

The Information Technology Department is responsible for the company's computers, servers, network infrastructure, software, cybersecurity, backups, and technical support.

Employee Distribution

Department	Employees	Primary Responsibilities
Information Technology	5	Software development, infrastructure, technical support
Human Resources	4	Recruitment, employee records, administration
Finance	5	Accounting, payroll, budgeting, financial records
Sales	6	Sales, customer relations, marketing
Total	20	

Enterprise Hardware Inventory

The hardware inventory was designed according to the company's 20 employees and departmental requirements. Each employee requires access to a computer, while specialized equipment such as servers, storage devices, UPS units, and networking equipment supports the entire organization.

Asset ID	Hardware	Quantity	Department	Purpose
HW-001	Desktop	15	HR, Finance, Sales	Daily office productivity and business applications
HW-002	Laptop	5	IT	Software development, administration, troubleshooting, and mobility
HW-003	Server	1	IT	Centralized services, file storage, applications, and internal systems
HW-004	Router	1	IT	Connects the internal network to the internet
HW-005	Managed Network Switch	1	IT	Connects wired devices and manages network traffic
HW-006	Network Printer	2	HR, Finance	Printing reports, documents, forms, and records
HW-007	UPS	4	IT	Provides backup power and protects critical equipment
HW-008	Wireless Access Point	3	IT	Provides wireless network connectivity
HW-009	NAS Storage	1	IT	Centralized file storage and backup
HW-010	External Backup Drive	2	IT	Offline backup and disaster recovery
HW-011	Monitor	20	All Departments	Primary display for employee computers

Hardware Justification

Fifteen desktop computers are assigned to HR, Finance, and Sales because these departments primarily perform office-based tasks. Five laptops are assigned to IT personnel because IT employees may need to move between workstations, network equipment, server rooms, and meeting areas.

A dedicated server is included to provide centralized infrastructure services. A managed switch is required to connect wired computers, printers, servers, and other network devices. Wireless access points provide wireless connectivity throughout the office.

The NAS provides centralized storage and supports the company's backup strategy. External backup drives are included so that important backups can also be stored separately from the primary network infrastructure.

UPS units are especially important for the server, NAS, networking equipment, and other critical devices because unexpected power interruptions could cause data loss or hardware damage.

Enterprise Software Inventory

Software	Version	License	Purpose
Windows 11 Pro	Current supported version	Commercial	Operating system for employee computers
Ubuntu Server	LTS Release	Open source	Server operating system
Microsoft Office	Microsoft 365	Subscription	Documents, spreadsheets, presentations, and email-related productivity
Visual Studio Code	Current stable version	Free	Software development and code editing
Git	Current stable version	Open source	Version control
Github Desktop	Current stable version	Free	Graphical Git repository management
VirtualBox	Current stable version	Free/open source	Virtual machine testing and development
Google Chrome	Current stable version	Free	Web browsing and web-based applications
Microsoft Defender	Included with windows	Included	Malware and endpoint protection
Anydesk	Current version	Commercial/free depending on use	Remote technical support
7-Zip	Current Stable version	Open source	File compression and archive management

Software Justification

Windows 11 Pro will be used on employee computers because it provides a professional desktop environment suitable for business applications and organizational management.

Ubuntu Server will serve as the operating system for the company's server because it is stable, widely used, and suitable for server administration.

Microsoft Office is required for creating documents, spreadsheets, presentations, reports, financial documents, and other business materials.

Visual Studio Code is required by IT personnel for software development and configuration work. Git and GitHub Desktop support source-code version control and collaboration.

VirtualBox allows IT employees to create virtual machines for testing operating systems, applications, and configurations without affecting production systems.

Google Chrome provides access to websites, cloud applications, development tools, and online services.

Microsoft Defender provides endpoint protection against malware and other common security threats.

AnyDesk can be used by authorized IT personnel for remote technical support when appropriate security controls are followed.

7-Zip provides file compression and extraction capabilities that are useful for managing backups, software packages, and archived files.

Enterprise Network Inventory

Asset ID	Network Equipment	Quantity	Purpose
NET-001	ISP Modem	1	Provides the internet connection from the ISP
NET-002	Router	1	Routes traffic between the office network and internet
NET-003	Firewall	1	Protects the internal network from unauthorized traffic
NET-004	Managed switch	1	Connects wired computers and network devices
NET-005	Wireless access point	3	Provides wireless connectivity
NET-006	Patch Panel	1	Organizes and terminates network cables
NET-007	CAT6 Ethernet Cable	30 rolls/lengths as required	Wired network connections
NET-008	RJ45 Connectors	100	Network cable termination and maintenance

Proposed Network Addressing

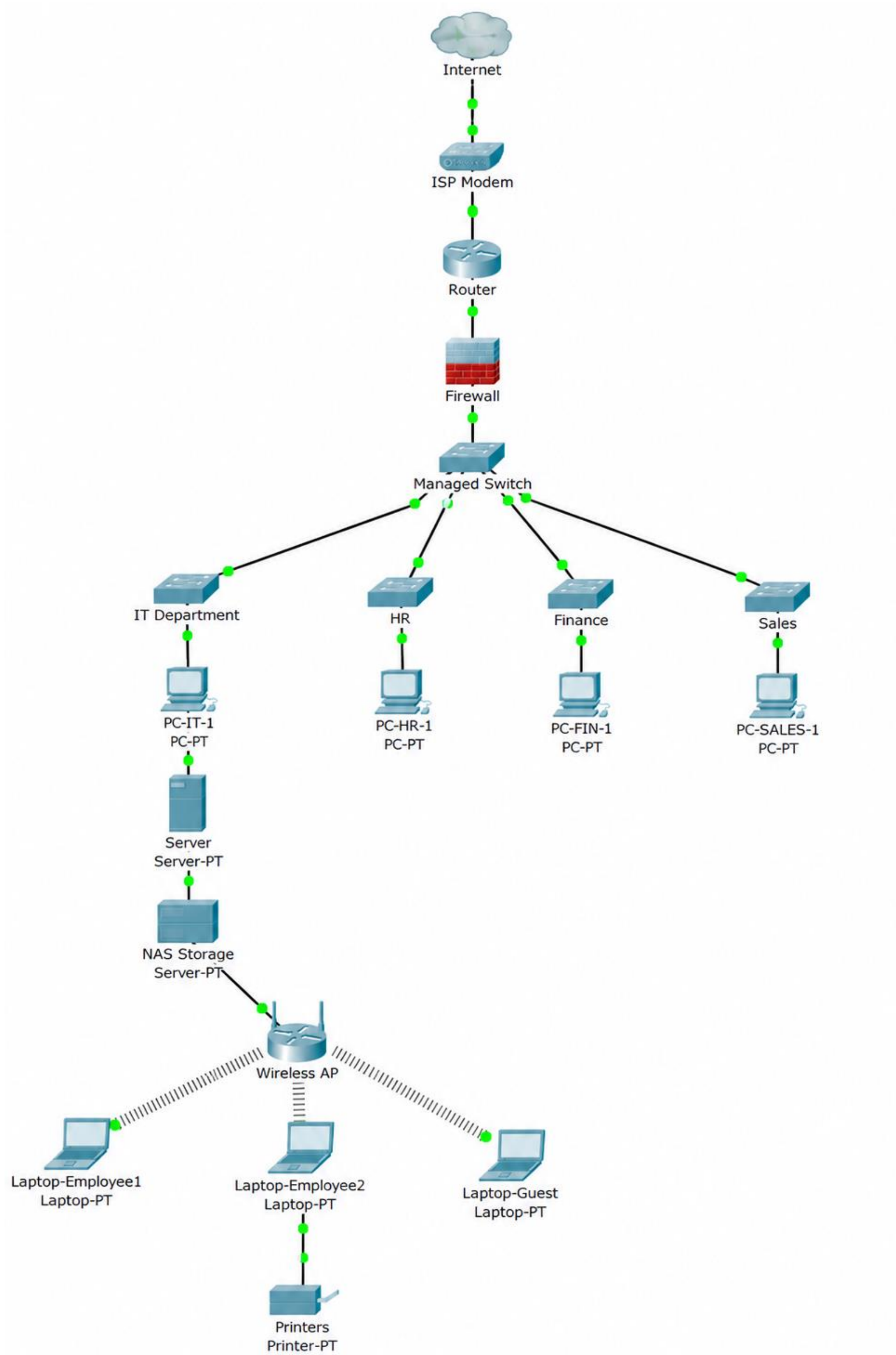
92.168.10.0/24

Suggested addressing:

Device/Department	Network
IT Department	192.168.10.0/26
HR Department	192.168.10.64/27
Finance Department	192.168.10.96/27
Sales Department	192.168.10.128/27
Servers/ Infrastructure	192.168.10.160/28
Network Management	192.168.10.176/28
Guest Wi-Fi	192.168.10.192/27

This addressing scheme provides logical separation between departments and leaves room for future expansion.

Enterprise Network Diagram



Network Design Explanation

The ISP modem provides the connection from the internet service provider. The router manages network routing, while the firewall provides an additional security layer between the public internet and the company's internal network.

The managed switch acts as the central connection point for wired devices. Computers from the IT, HR, Finance, and Sales departments connect to the switch. The server and NAS are also connected to the internal network.

Wireless access points provide wireless connectivity for authorized employees and guests. Guest wireless access should be isolated from internal company systems.

The network should be designed in Draw.io using standard networking symbols and clearly labeled connections. The final diagram should be exported in both PNG and PDF formats as required by the project instructions.

System Administration Roles

<u>Helpdesk Technician</u>

• Responsibilities

A Helpdesk Technician provides first-level technical support to employees. Typical responsibilities include troubleshooting computer problems, installing software, resolving login issues, assisting users with network connectivity, documenting incidents, and escalating complex problems to other IT personnel.

• Skills

Important skills include troubleshooting, communication, customer service, operating system knowledge, hardware knowledge, basic networking, documentation, and problem-solving.

• Common Tools

Common tools include ticketing systems, remote-support applications, Windows administrative tools, antivirus software, hardware diagnostic tools, and knowledge bases.

• Certifications

Possible certifications include CompTIA A+, Microsoft-related certifications, and other entry-level IT support certifications.

<u>Network Administrator</u>

• Responsibilities

A Network Administrator manages and maintains the organization's network infrastructure. Responsibilities include configuring routers and switches, managing IP addresses, monitoring network performance, troubleshooting connectivity problems, maintaining wireless networks, and implementing network security controls.

• Skills

A Network Administrator should understand TCP/IP, subnetting, routing, switching, DNS, DHCP, wireless networking, firewalls, network monitoring, and cybersecurity.

• Common Tools

Common tools include network monitoring software, command-line utilities, router and switch management interfaces, Wireshark, ping, traceroute, and firewall management tools.

• Certifications

Possible certifications include Cisco CCNA, CompTIA Network+, and other networking certifications.

<u>Linux System Administrator</u>

• Responsibilities

A Linux System Administrator manages Linux-based servers and services. Responsibilities include installing and configuring Linux systems, managing users and permissions, monitoring system performance, managing storage, installing updates, troubleshooting services, and maintaining server security.

• Skills

Important skills include Linux command-line administration, shell scripting, file permissions, networking, storage management, system monitoring, package management, and troubleshooting.

• Common Tools

Common tools include SSH, Bash, systemctl, journalctl, top, df, ip, apt, and other Linux command-line utilities.

• Certifications

Possible certifications include Linux+, LPIC-1, and Red Hat certifications.

<u>Cloud Administrator</u>

• Responsibilities

A Cloud Administrator manages cloud-based infrastructure and services. Responsibilities include deploying virtual machines, managing cloud storage, configuring networks, managing user access, monitoring cloud resources, implementing security controls, and controlling cloud costs.

• Skills

A Cloud Administrator should understand virtualization, cloud networking, identity and access management, storage, security, backups, monitoring, and cloud architecture.

• Common Tools

Common tools include cloud provider management consoles, command-line interfaces, monitoring tools, identity-management systems, and infrastructure-management tools.

• Certifications

Possible certifications include Microsoft Azure Administrator, AWS Certified Cloud Practitioner, AWS Certified Solutions Architect, or similar cloud certifications.

Collaboration Between the Four Professionals

The four professionals work together to maintain a reliable IT environment. The Helpdesk Technician serves as the first point of contact for employees and resolves common technical problems. When an issue involves network connectivity, the Network Administrator investigates the network infrastructure. If the problem involves Linux servers or server-based services, the Linux System Administrator handles the server-side investigation. The Cloud Administrator manages cloud-based resources and services when the organization depends on cloud infrastructure. Effective communication and documentation between these professionals allow technical problems to be resolved efficiently while reducing downtime and maintaining business operations.

Infrastructure Recommendations

• Internet Providers

Euphoria Solutions should subscribe to a reliable business-grade internet service with sufficient bandwidth for 20 employees. A minimum target of approximately 300 Mbps to 500 Mbps is reasonable for the initial deployment, depending on actual business usage and available local providers.

A business-grade connection is preferred because the company depends on internet access for software development, cloud services, communication, remote support, online collaboration, and customer services.

The company should also evaluate whether the provider offers a service-level agreement and reliable technical support.

• Server Specifications

The initial server should have enough capacity to support current services while allowing future growth.

Recommended starting specifications:

- 8-core server-class processor
- 32 GB ECC RAM
- 2 × 1 TB SSD in RAID 1 for the operating system and important services
- Additional storage for application and file requirements
- Dual network interfaces
- Redundant or reliable power supply

- UPS protection
- Ubuntu Server LTS
- Automated monitoring and backup

The server should not be treated as the only location for important data. Critical information should also be backed up to the NAS and external backup media.

• Backup Strategy

Euphoria Solutions should follow a 3-2-1 backup strategy:

- Keep at least three copies of important data.
- Store the copies on at least two different types of storage.
- Keep at least one backup copy offline or in a separate location.

The primary data can be stored on the server, with a secondary backup on the NAS. External backup drives can provide an additional offline copy.

Backups should be scheduled automatically and tested regularly. A backup that has never been tested should not be considered fully reliable.

• Security Recommendations

The company should implement security controls from the beginning because the infrastructure is being built from scratch.

Recommended controls include:

- Firewall protection
- Strong password requirements
- Multi-factor authentication where available
- Regular operating system updates
- Software patch management
- Endpoint protection
- User access controls
- Department-based network segmentation
- Guest Wi-Fi isolation
- Regular backups
- Physical security for network equipment
- Restricted administrative privileges
- Security awareness training
- Regular review of user accounts

• Antivirus and Endpoint Protection

Microsoft Defender should be enabled on supported Windows computers. Security software should remain updated and regularly monitored.

Employees should also be trained not to open suspicious attachments, install unauthorized software, or provide their credentials to unknown individuals.

• Password Policy

The company should establish a formal password policy requiring employees to use strong and unique passwords.

Recommended practices include:

- Use long passwords or passphrases.
- Do not reuse passwords between systems.
- Do not share passwords with other employees.
- Enable multi-factor authentication when supported.
- Administrative accounts should be separate from normal user accounts.
- Passwords should never be stored in plain text.
- Compromised passwords must be changed immediately.

• Expansion Plan

The infrastructure should be designed with future growth in mind. The current network should have spare switch ports, additional IP address capacity, and sufficient server and storage resources.

If the company grows from 20 employees to 40 or 50 employees, additional switches, wireless access points, computers, storage, and possibly servers can be added.

The company should also consider virtualization and cloud services as its infrastructure requirements increase.

Reflection

This project helped me understand that system administration is not only about fixing computers after something goes wrong. A System Administrator must also understand the organization, identify its requirements, plan the infrastructure, select appropriate hardware and software, design the network, and document technical decisions before deployment. Through this project, I learned how different parts of an IT infrastructure work together to support business operations.

One of the most important things I learned was how to create an inventory based on actual business requirements. Instead of simply listing computer equipment, I had to determine which devices were appropriate for each department and explain why they were needed. I also learned that servers, network equipment, storage, UPS devices, printers, and wireless access points all have specific purposes within an organization.

The most challenging task for me was designing the enterprise network topology. I needed to consider how the internet, modem, router, firewall, switch, server, wireless access points, printers, and different departments should be connected. Creating a logical network structure helped me understand how network devices work together and why proper planning is important.

Planning is important before deployment because mistakes made during the planning stage can result in unnecessary expenses, security problems, poor performance, and difficult maintenance later. A properly planned infrastructure can provide better reliability, security, scalability, and organization. It also gives administrators a clear reference when troubleshooting or expanding the network.

This project will help me become a better System Administrator because it allowed me to practice thinking about IT infrastructure from an organizational perspective rather than focusing only on individual computers. I learned that good system administration requires technical knowledge, documentation, communication, organization, and critical thinking. In the future, I can apply these skills when designing and maintaining real-world IT environments.